

PORTFOLIO

Creative

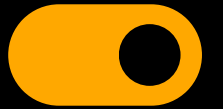
Creative Portfolio Presentation

Presented By: Rizqi Akbar Maulana, S.Kel



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ABOUT ME

I'm a Marine Science graduate with a strong focus on spatial data analysis, remote sensing, and marine GIS. I have hands-on experience in processing satellite imagery, visualizing oceanographic parameters, and transforming complex environmental data into clear, actionable insights. Through academic projects, assistantship roles, and fieldwork, I have developed solid technical skills in GIS, Python, and scientific visualization, which I apply to support research and data-driven decision making.

EDUCATION

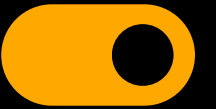


2021 - 2025

Marine Science – Padjadjaran University

GPA: 3.51/4.00



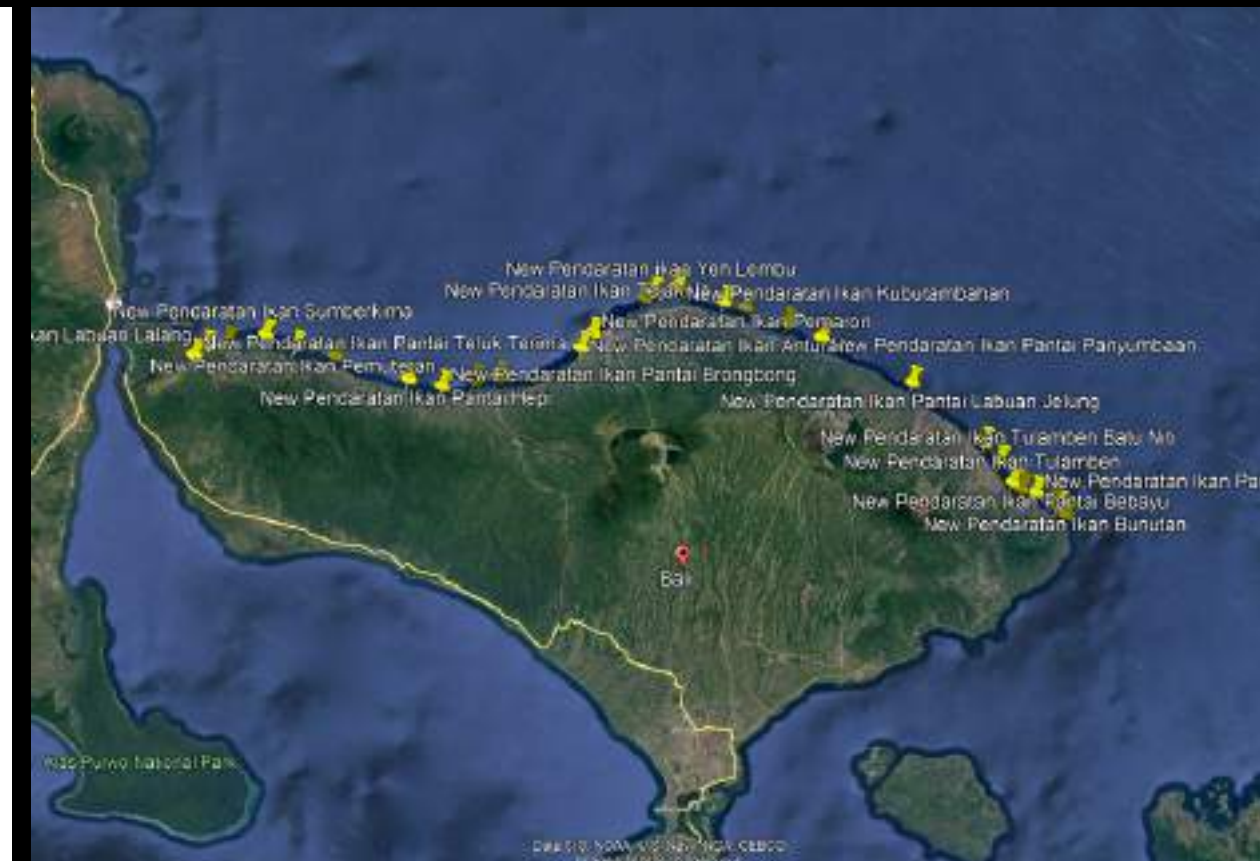
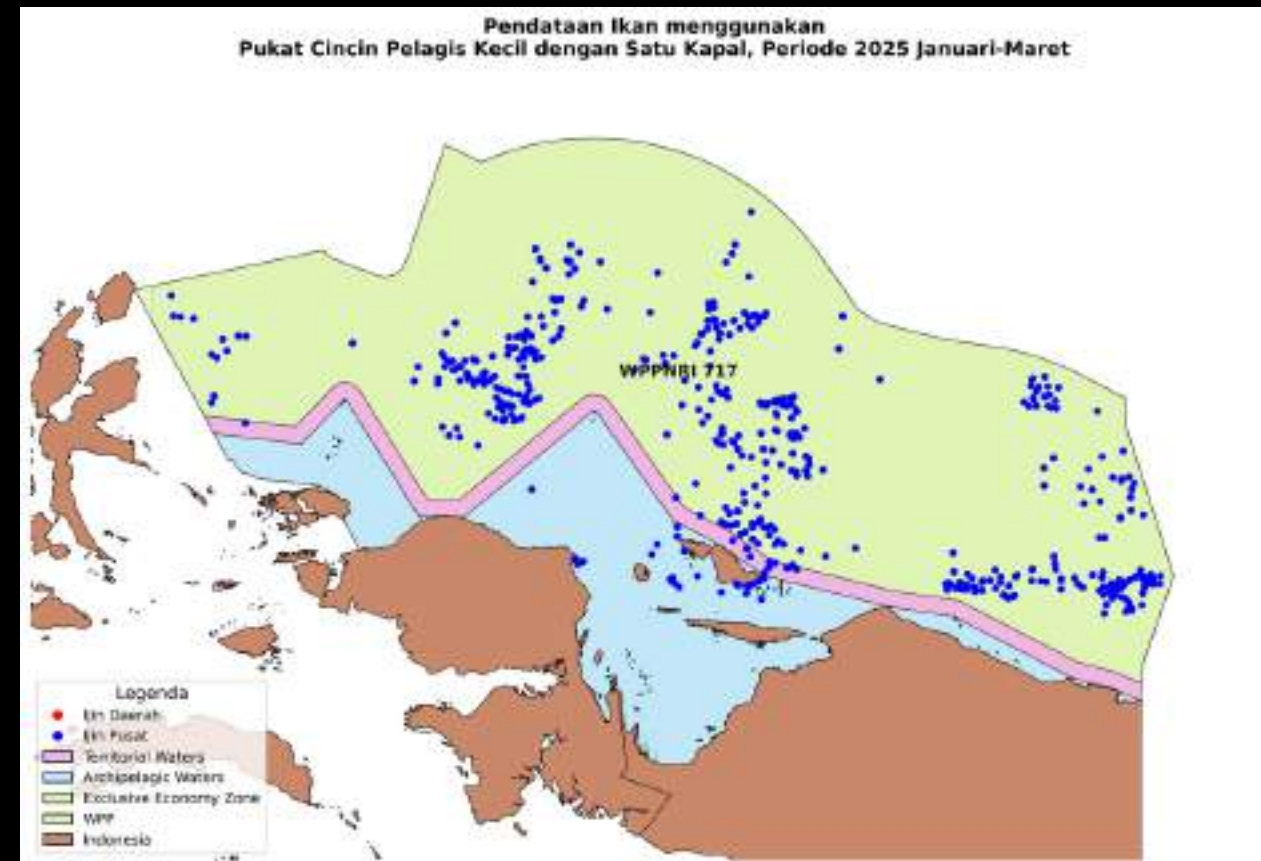


SOFTWARE SKILLS



PERSONAL SKILLS

- Communication & Public Speaking
- Team Collaboration
- Leadership
- Problem Solving
- Critical Thinking
- Time Management
- Adaptability
- Attention to Detail
- Project & Fieldwork Management
- Decision-making



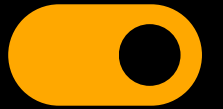
WORK EXPERIENCE

DEC 25 - JUN 2026

Ministry of Marine Affairs and Fisheries

- Produced over 50 thematic fish landing maps covering the entire National Fisheries Management Area of the Republic of Indonesia (FMA)
- Produced 9 site plans for the location of Merah Putih Fishing Villages (KNMP) based on field survey results
- Collecting data on fish landings at fishing centers throughout the FMA
- Cleaning and validating approximately 637,000 fishery data records
- Creating shapefiles of maritime boundaries, including Indonesia's Archipelagic Waters (AW), Territorial Waters (TW), and Exclusive Economic Zone (EEZ)



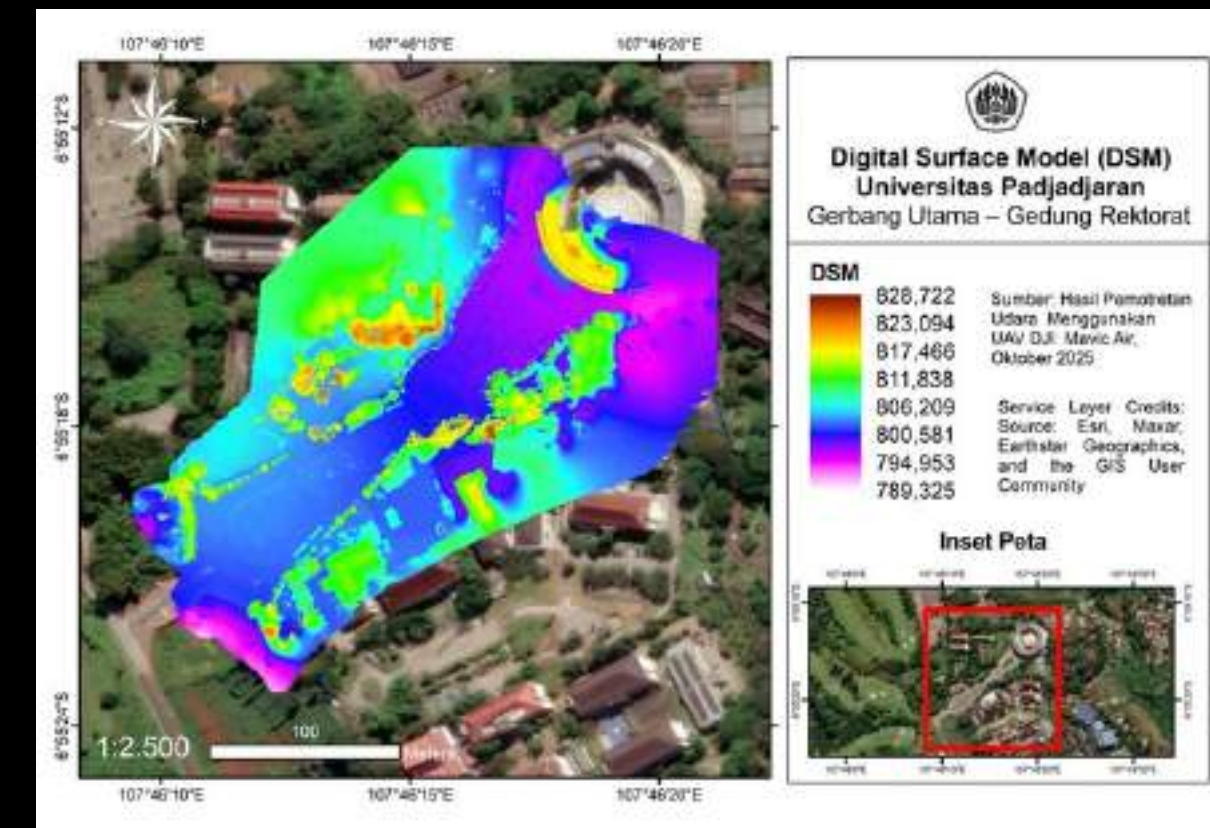


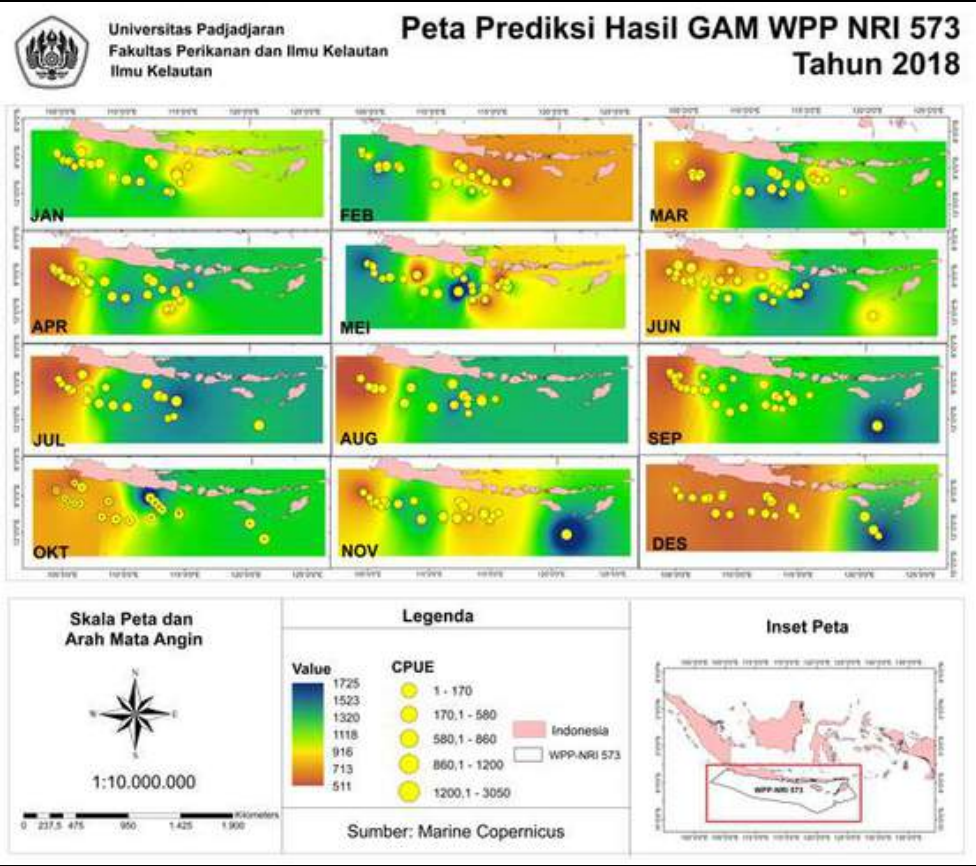
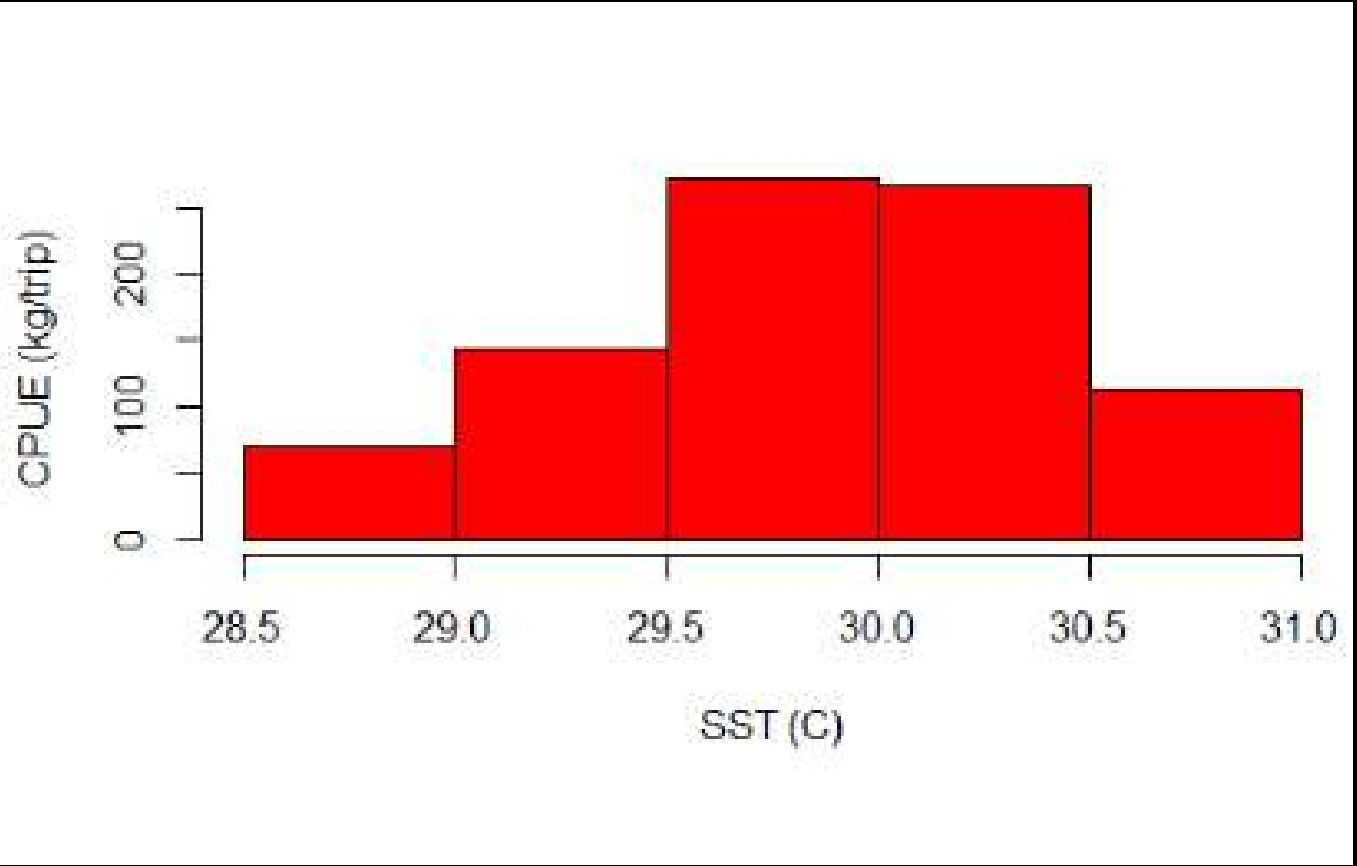
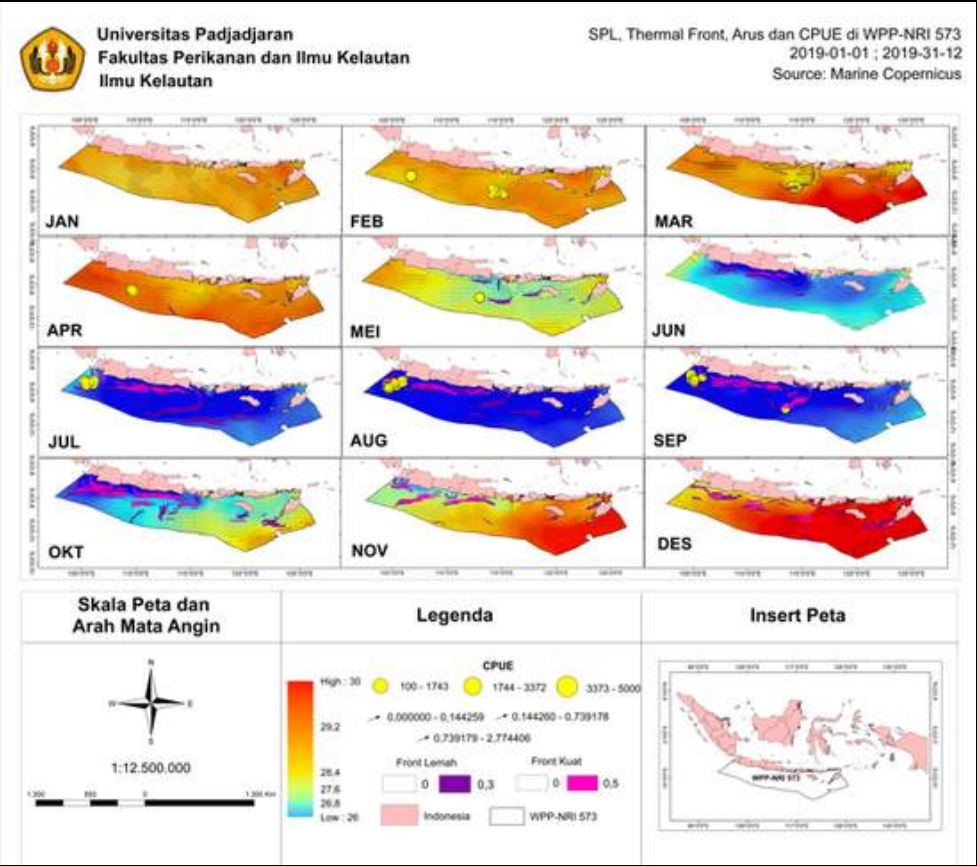
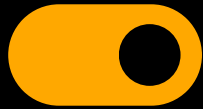
WORK EXPERIENCE

OCT - NOV 2025

Coastal Geomorphology Practicum Assistant

- Developing and teaching more than 4 practical modules related to coastal slope analysis using drones and waterpasses.
- Guiding students in the process of field data acquisition and technical and methodological interpretation of coastal slopes.
- Directing the use of ArcGIS, Excel, and Agisoft software for data processing, spatial analysis, and visualization of results.





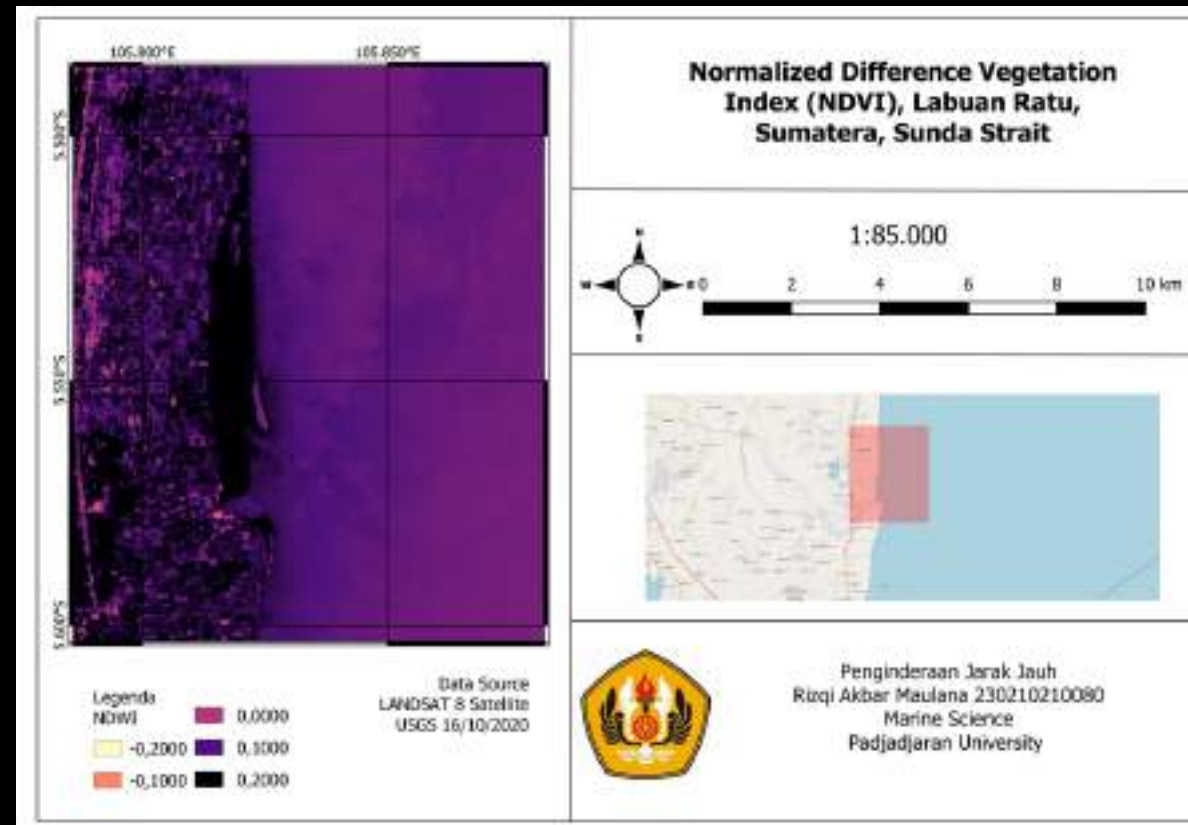
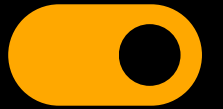
WORK EXPERIENCE

SEP - NOV 2025

Fisheries Oceanography Practicum Assistant

- Developing and teaching 8+ practical modules related to hydro-oceanographic and spatial data analysis and the use of GAM models.
- Guiding students in obtaining and interpreting satellite images (SPL, salinity, chlorophyll-a, sea level height, wind, currents, and front formation estimates).
- Guiding students in obtaining and interpreting fish catch data.
- Directing the use of ArcGIS, Excel, and Rstudio software for data visualization and processing.



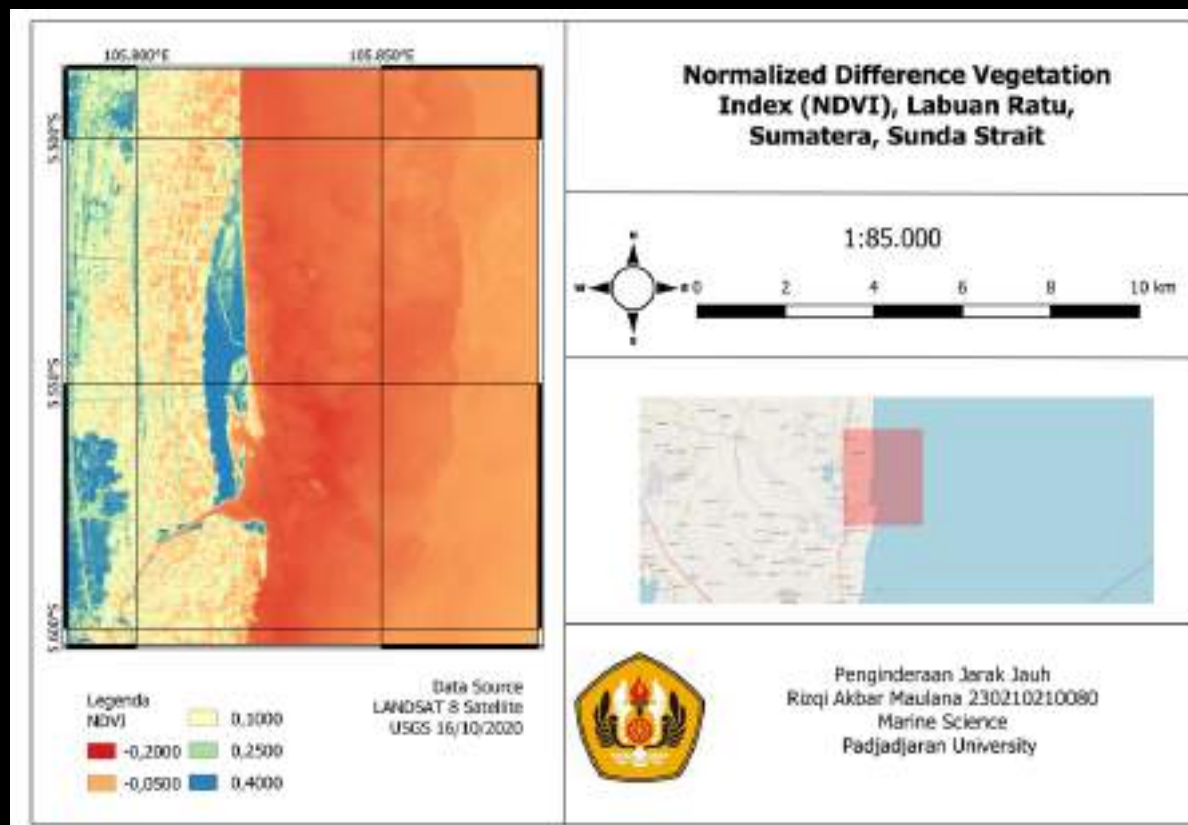


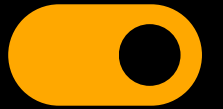
WORK EXPERIENCE

MAR - JUN 2025

Remote Sensing Practicum Assistant

- Developing and teaching 4+ practical modules related to hydro-oceanographic and spatial data analysis.
- Guiding students in obtaining and interpreting satellite imagery (SPL, salinity, chlorophyll-a, coastline changes).
- Directing the use of ArcGIS, QGIS, Python, and Jupyter Notebook software for data visualization and processing.





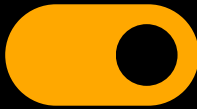
WORK EXPERIENCE

SEP - NOV 2024

Operational Staff – Bening Saguling Foundation



- Supporting water quality monitoring activities in the Citarum River at more than 10 sampling points.
- Coordinating field operations and compiling environmental program documentation.
- Serving as Master of Ceremony (MC) and speaker in visit and outreach agendas with government agencies.



WORK EXPERIENCE

MAR - JUN 2024

Computer and Big data Practicum Assistant

- Designing basic Python learning modules for data analysis and visualization.
- Guiding students in applying Python to data-based projects, including processing, analysis, and interpretation of results.

```
import pandas as pd
import matplotlib.pyplot as plt

# Membaca data dari file excel
data_karang = pd.read_excel("C:\\Users\\arisaq\\Documents\\prak kmbig anak 23\\Praktikum.xlsx")

# Mengatur agar tampilan untuk menampilkan semua baris dan kolom
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)

# Menampilkan ukuran dan seluruh data dari dataframe
print(data_karang.shape)
print(data_karang)

# Mengambil data dari kolom yang diperlukan
tahun = data_karang['Tahun'].values
suhu = data_karang['Suhu'].values
kerapatan = data_karang['Kerapatan karang'].values

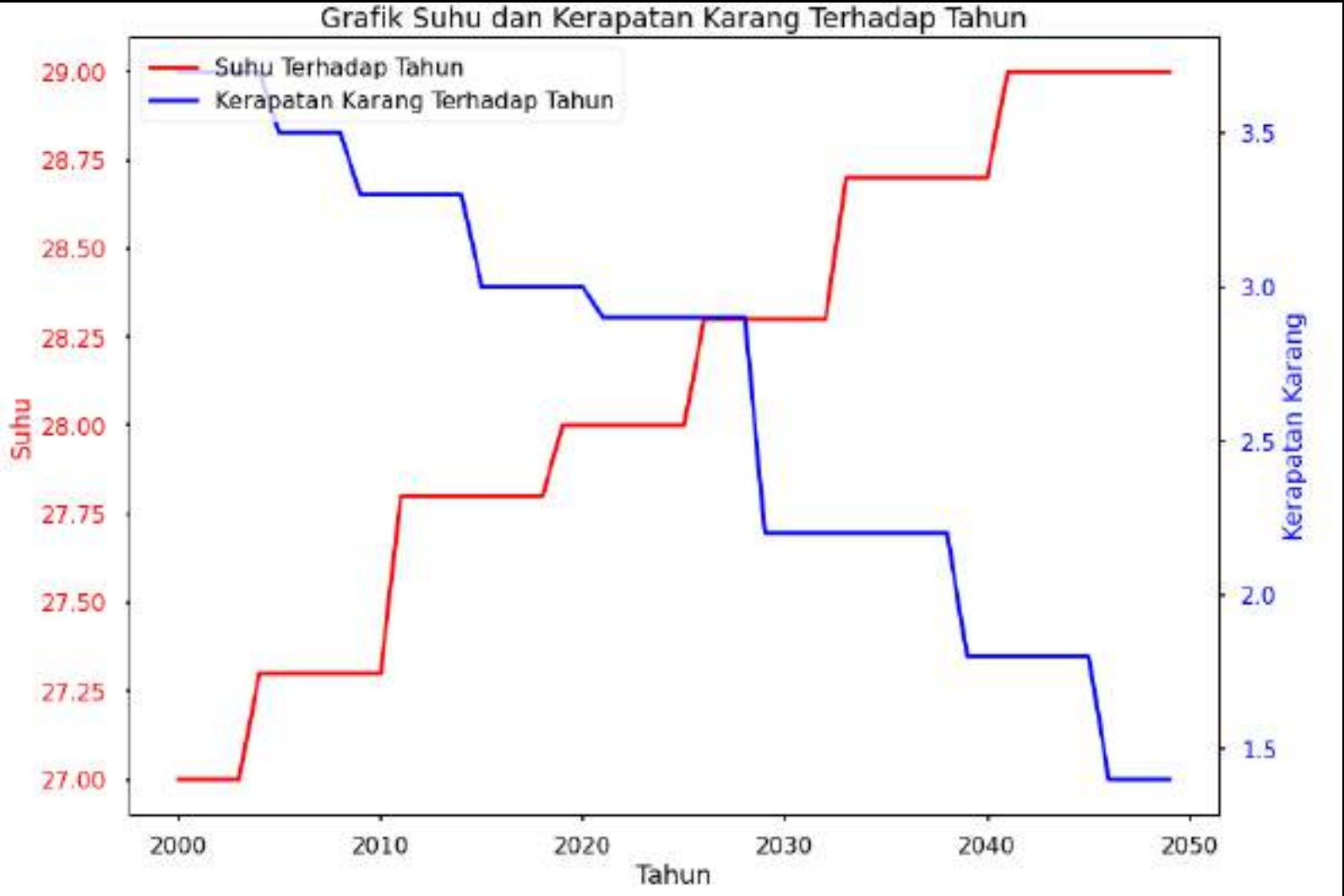
# Membuat plot
fig, ax1 = plt.subplots()

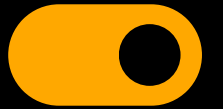
# Grafik suhu terhadap tahun
ax1.plot(tahun, suhu, label='Suhu Terhadap Tahun', color='r', linestyle='--')
ax1.set_xlabel('Tahun')
ax1.set_ylabel('Suhu', color='r')
ax1.tick_params(axis='y', labelcolor='r')

# Membuat axis kedua untuk kerapatan karang
ax2 = ax1.twinx()
ax2.plot(tahun, kerapatan, label='Kerapatan Karang Terhadap Tahun', color='b', linestyle='--')
ax2.set_ylabel('Kerapatan Karang', color='b')
ax2.tick_params(axis='y', labelcolor='b')

# Menambahkan judul dan legend
plt.title('Grafik Suhu dan Kerapatan Karang Terhadap Tahun')
fig.tight_layout()
fig.legend(loc='upper left', bbox_to_anchor=(0,1), bbox_transform=ax1.transAxes)

# Menampilkan grafik
plt.show()
```

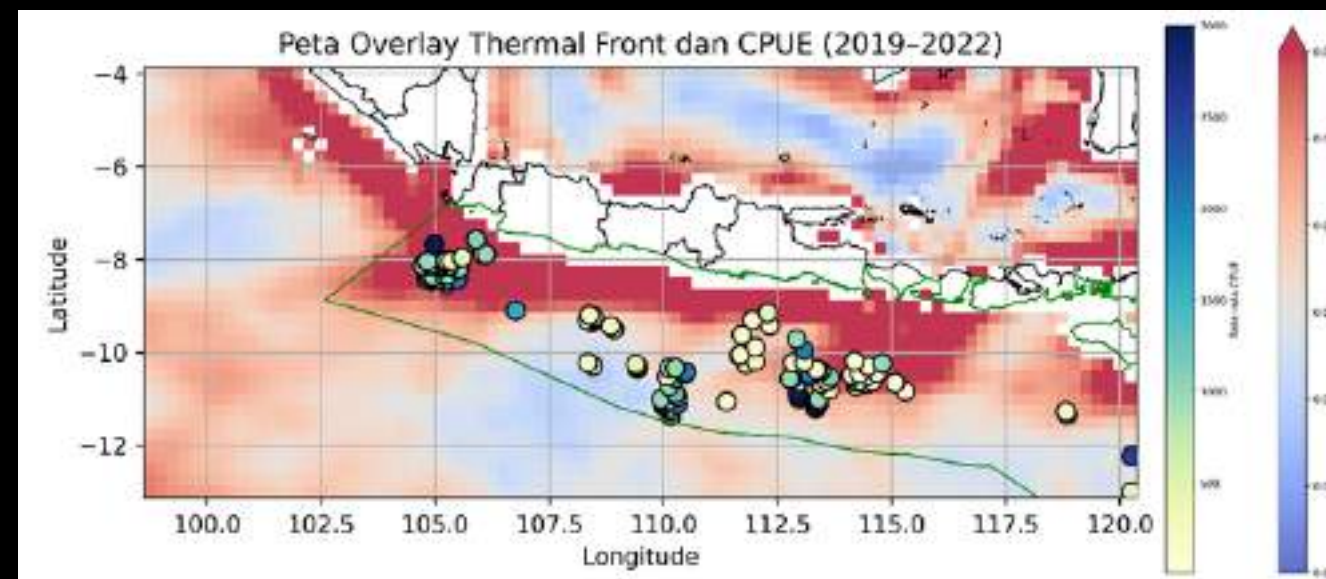
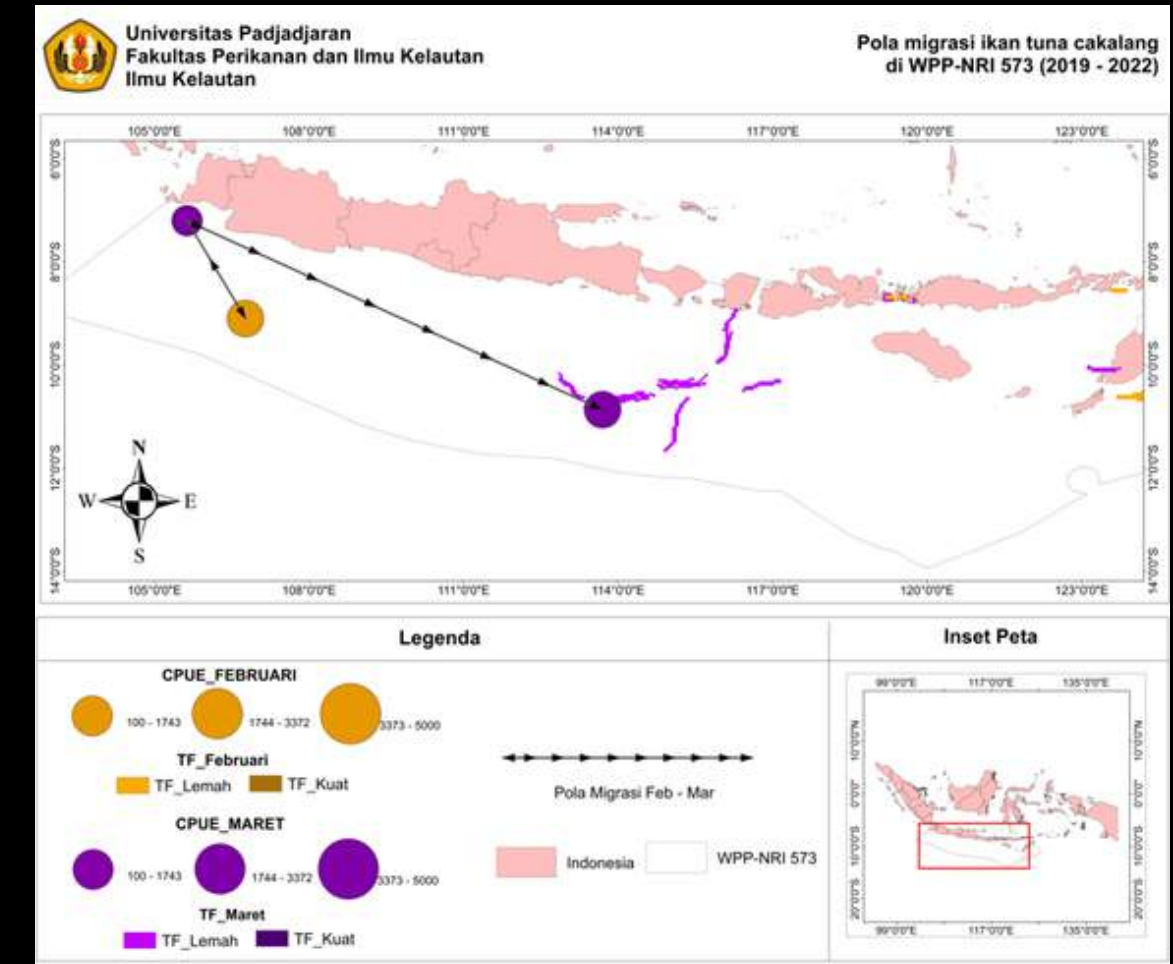
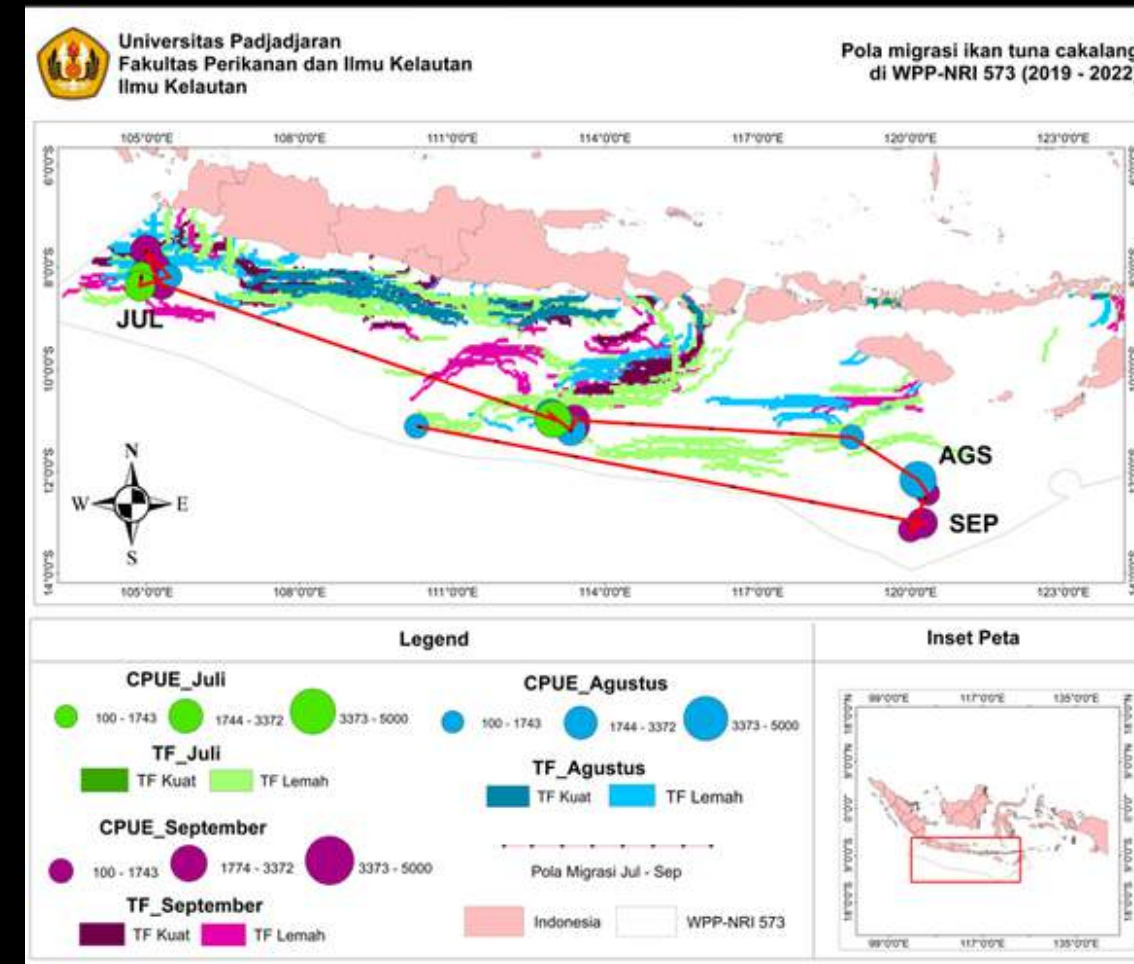


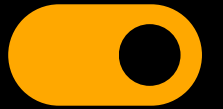


WORK EXPERIENCE

Thesis : Thermal front analysis and its effect on the migration patterns of skipjack tuna (*Katsuwonus pelamis*) in WPP-NRI 573

This thesis analyzes thermal fronts and their influence on skipjack tuna (*Katsuwonus pelamis*) migration in WPP-NRI 573. Using satellite-derived SST data and spatial analysis, the study examines the formation and seasonal variation of thermal fronts and relates them to tuna movement patterns. The results support better fisheries management and more accurate identification of potential fishing grounds.



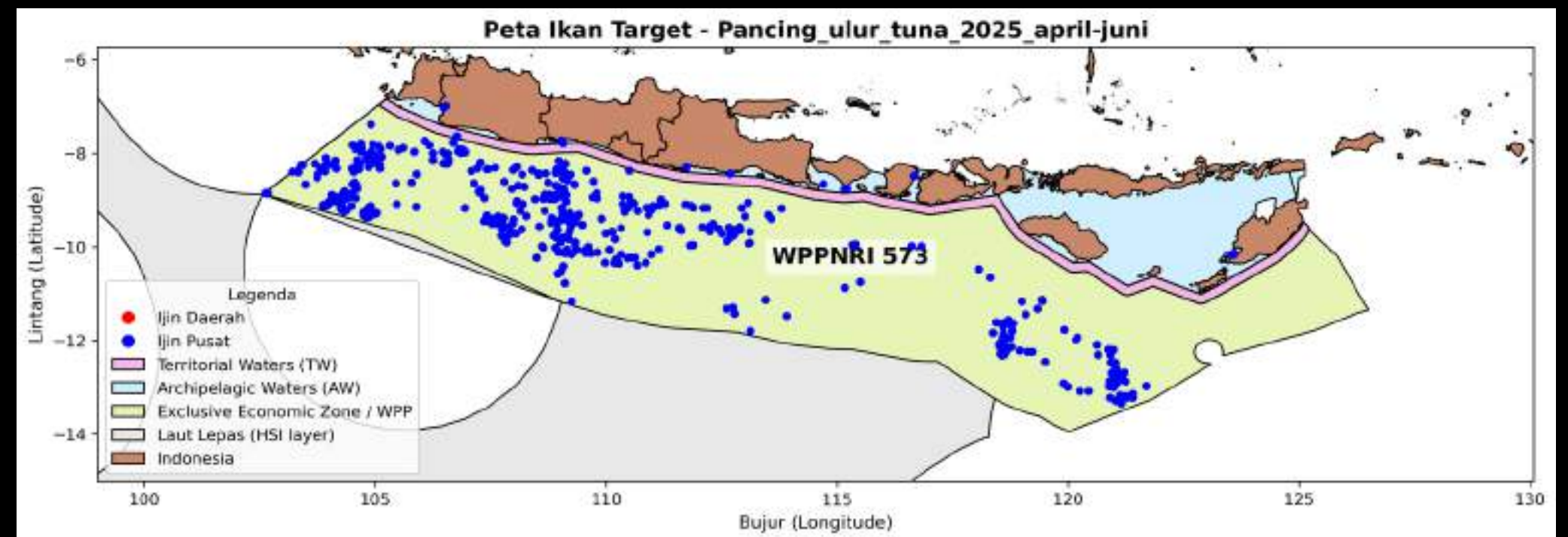
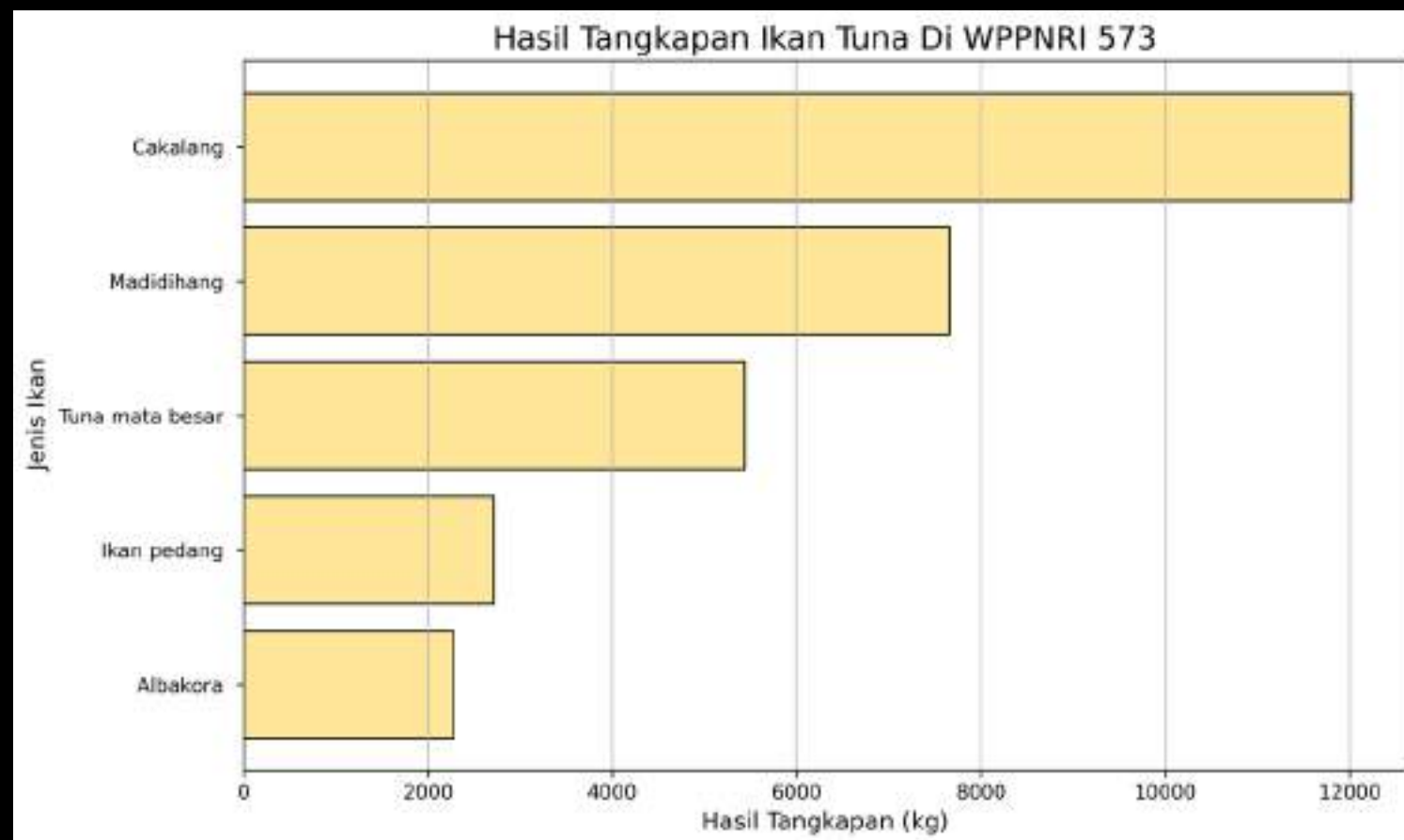


SPOTLIGHT PROJECT

DEC 2025 - JUN 2026

Mapping & Analysis of Fishing Data – KKP

This project analyzes the spatial distribution of tuna catches in WPP-NRI 573 based on KKP catch logbook data. Water zone layers (EEZ, Archipelagic Waters, Territorial Waters) were constructed from official reference polylines using ArcGIS and Python, then used as the basis for catch point classification. The output includes 10+ thematic maps per fishing gear and graphs of catch composition per fish species — skipjack tuna dominates with the highest catch compared to madidihang, bigeye tuna, swordfish, and albacore.





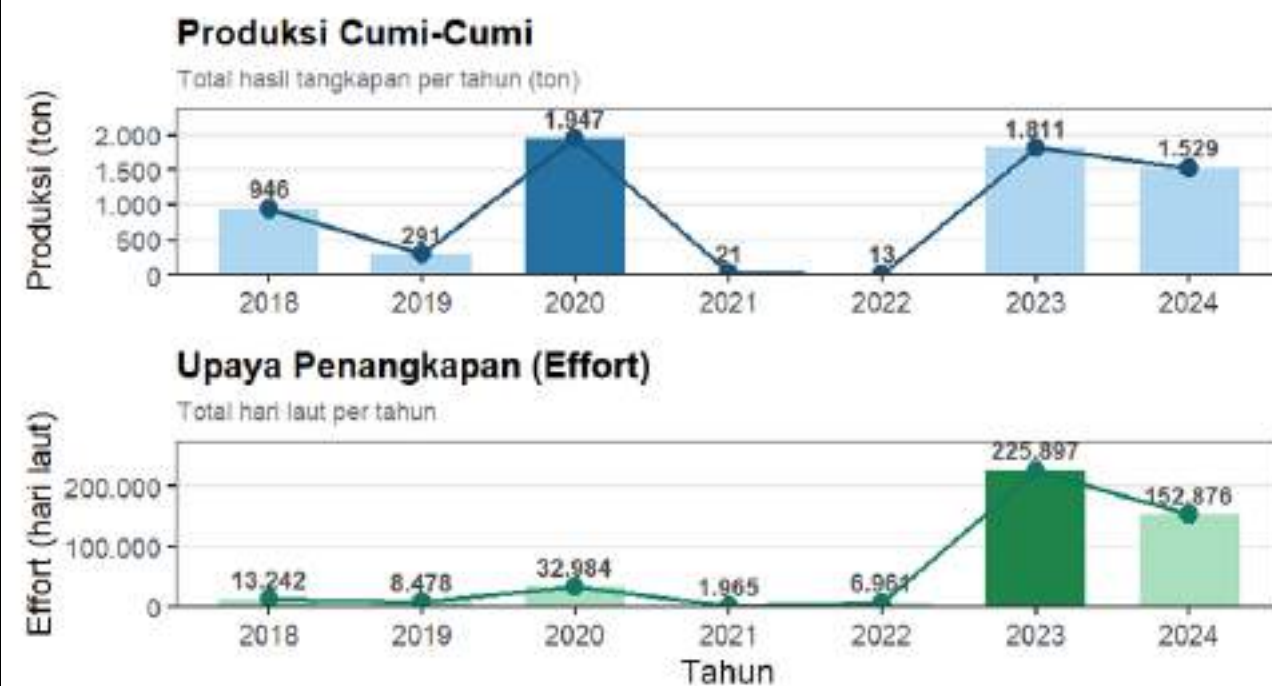
PROJECT HIGHLIGHT

R STUDIO - DATA: LOGBOOK KKP (2018-2024)

Squid Stock Assessment FMA 573 (Independent Project)

Tren Produksi & Upaya Penangkapan Cumi-Cumi

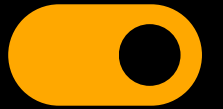
WPP 573 — Samudera Hindia Selatan Jawa | 2018–2024



- Production fluctuated sharply: highest 1,947 tons (2020), lowest 13 tons (2022)
- Effort increased dramatically in 2023 — reaching 225,897 sea days, up 32× from the previous year
- Increased effort was not matched by production → early indication of stock decline

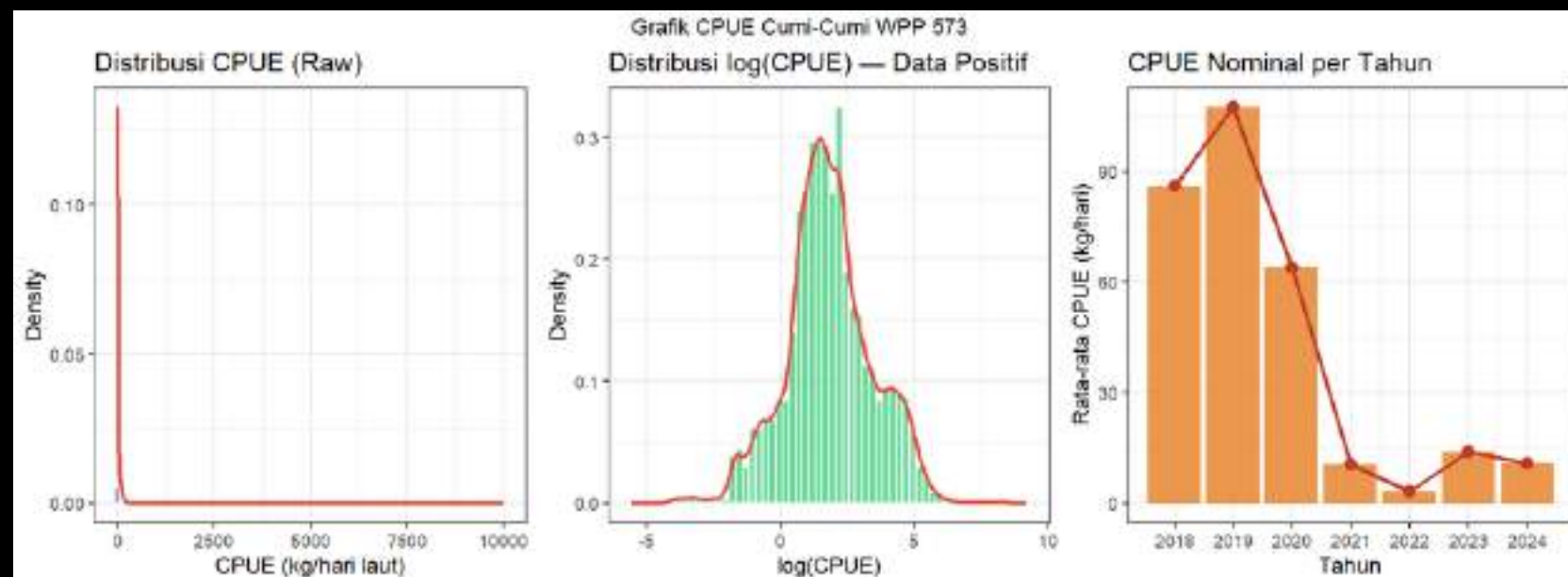
Tools : R Studio | ggplot2 | dplyr | sf



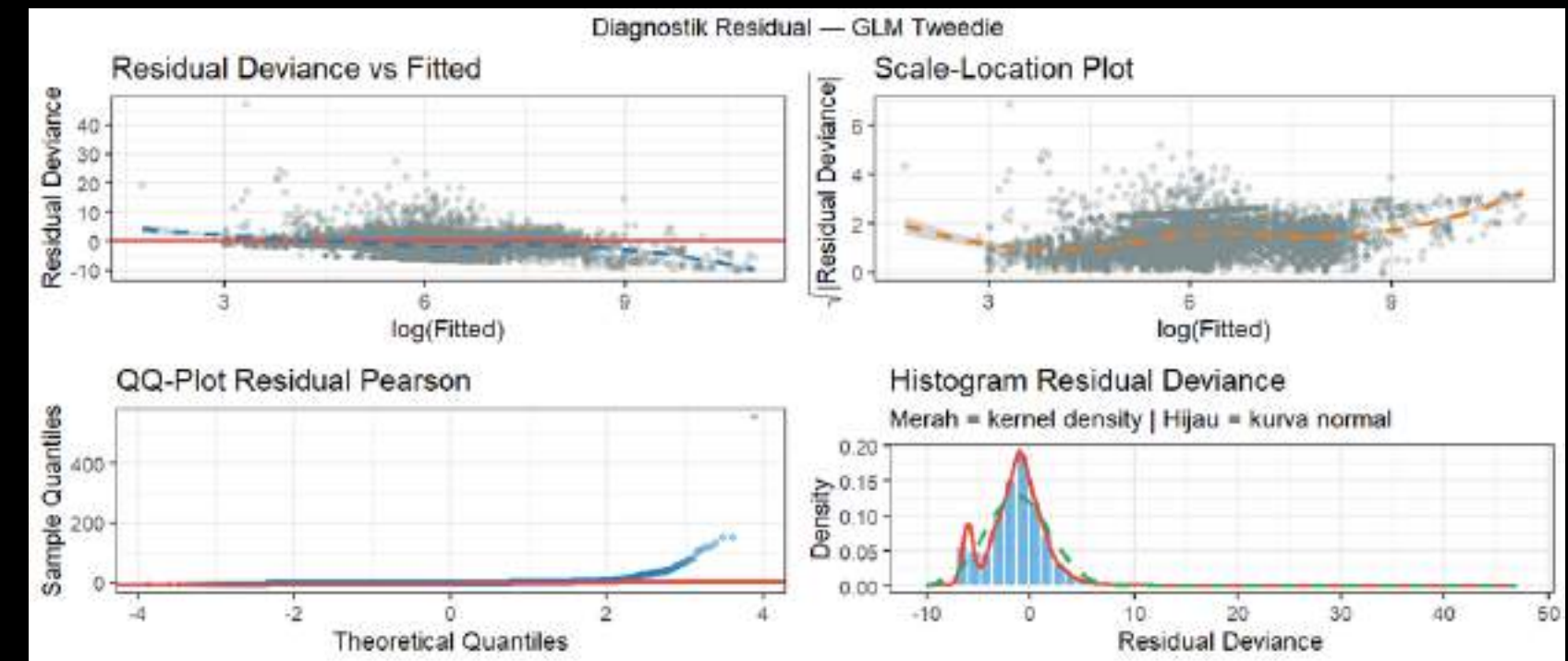


PROJECT HIGHLIGHT

Squid Stock Assessment FMA 573 (Independent Project)

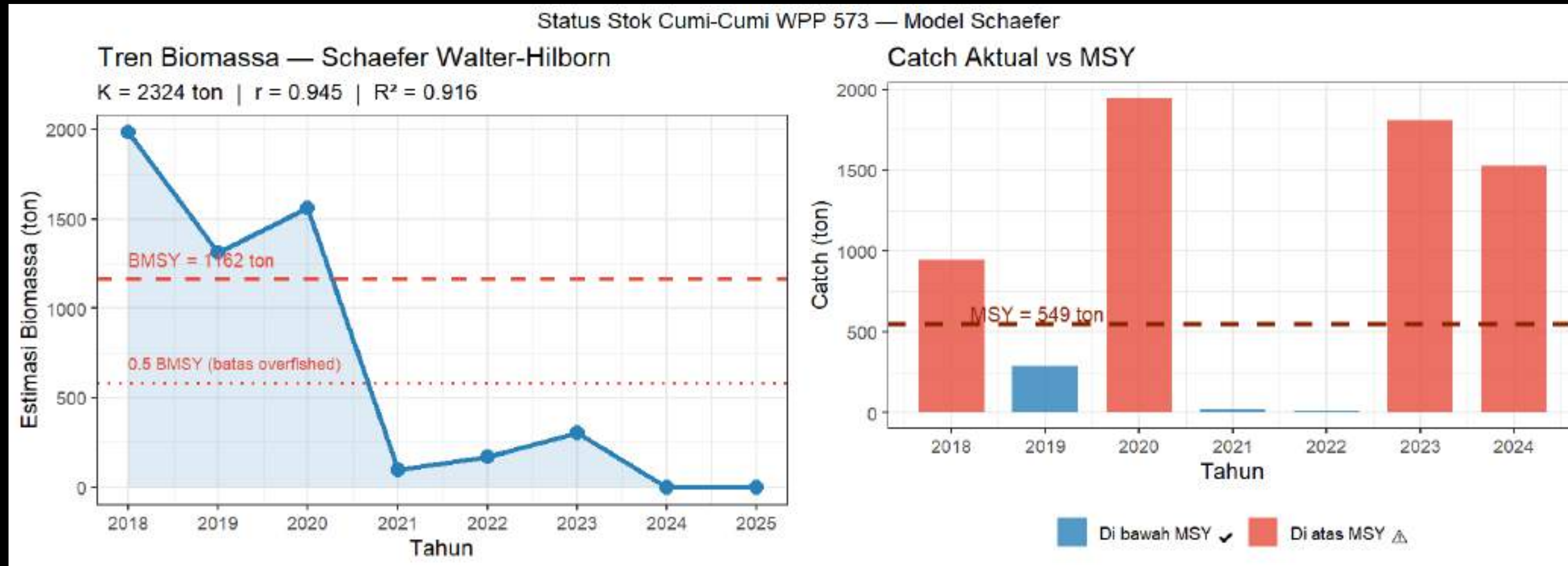
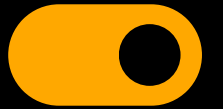


Raw CPUE is heavily skewed to the left—the majority of trips yield catches close to zero. $\log(\text{CPUE})$ transformation produces a near-normal distribution, validating the use of GLM for standardization. Nominal CPUE per year shows a sharp decline from ~88 kg/day (2018) to ~5 kg/day (2021–2024)



Residual Deviance vs Fitted shows a random pattern around zero — there is no systematic bias. The QQ-Plot shows a heavy tail, which is reasonable for catch data. The residual histogram is close to normal around zero. Overall, the model fit is quite good for data with many zeros.

Tools : R Studio | tidyverse | ggplot2 | gridExtra | minpack.lm



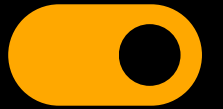
PROJECT HIGHLIGHT

Squid Stock Assessment FMA 573 (Independent Project)



Tools : R Studio | tidyverse | ggplot2 | gridExtra | minpack.lm

- Biomass declined dramatically from 2,000 tons (2018) to near zero in 2024–2025 — well below the BMSY (1,162 tons) and the overfished threshold (0.5 BMSY = 581 tons). The model fit is excellent: $R^2 = 0.916$.
- 5 out of 7 years of actual catches exceeded MSY (549 tons) — marked in red. Only 2019 and 2021–2022 were below MSY, and even then it was due to the pandemic rather than good stock management.



ORGANIZATION EXPERIENCE



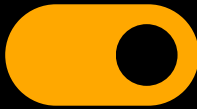
Vice Chair of the Student Executive Board, Faculty of Fisheries and Marine Sciences, Padjadjaran University, 2024/2025



General Coordinator Independent Marine Protected Animals Community (IMPACT) Unpad (Aug - Dec 2023)

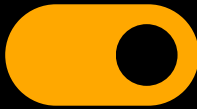


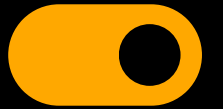
Head of External Affairs, Himaika Unpad 2023/2024



CERTIFICATION







GET IN TOUCH

If you're looking to collaborate with someone who thrives at turning data into clarity, transforming ideas into real impact, and bridging the gap between science and visual storytelling, I'd love to connect. Whether it's a research project, data analysis, environmental work, or creative mapping, feel free to reach out anytime. Let's build meaningful solutions, explore new possibilities, and create something that truly matters.

**Email**

arizqi828@gmail.com

**Social Media**

@akbrmlna08

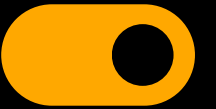
**LinkedIn**

<http://www.linkedin.com/in/rizqiakbar21002>

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THANK YOU

For Your Attention

